

The Business Case for Paying for Quality

A policy brief for dairy processors and large aggregators in Uganda

Based on "When Quality (doesn't) Pay: Evidence from Two Experiments in Ugandan Dairy" (Van Campenhout, Ariong, Kariuki, and Chamberlin). July 2026.

The problem in one sentence

Under flat, per-liter pricing, a processor pays exactly the same for a liter of clean, high-solids milk as for a liter that has been watered down or skimmed. That is not a farmer problem or a trader problem. It is a processor problem, because the cost lands on your yield, your product grade, and your margin.

Our two field experiments in Uganda's dairy sector show that this is fixable, and that fixing it is profitable rather than charitable. Making quality measurable at Milk Collection Centers (MCCs) cut serious adulteration roughly in half, and it did so without raising what farmers were paid. The quality was there to be had. What was missing was a buyer paying for it.

What the evidence shows

Measurement alone stops the bleeding. When we placed milk analyzers and digital record-keeping at collection centers (a cluster-randomized trial across 95 catchment areas), the share of deliveries containing more than 5 percent added water fell from 11 percent to 5 percent. Average added water dropped by about half a percentage point and butterfat rose by roughly 0.11 percentage points. Crucially, this happened through enforcement, the credible threat of being tested, not through paying farmers more. Prices did not move anywhere in the chain.

A price is what turns measurement into upgrading. In a second experiment we paid a per-liter quality bonus to traders. When we set a demanding quality bar, average butterfat in delivered milk rose by 0.17 percentage points. When the bar was easy to clear, traders simply delivered more volume (about 123 additional liters per day) and did not bother improving quality. The lesson for a buyer is precise: a quality standard only changes behavior when it actually bites.

Price simulation: what flat pricing is quietly costing you

The figures below are illustrative, calibrated to the effect sizes measured in the experiments and to Ugandan price points. They are meant to size the prize, not to replace your own accounting.

Simulation A. You are paying milk prices for water.

Assume a mid-size processor taking in 20,000 liters per day, 350 operating days per year, at an intake price of 1,200 Ugandan shillings (UGX) per liter (3,700 UGX per US dollar).

Item	Value
Average added water in intake (control)	1.63%
Water bought at milk price	~326 L/day
Cost of that water	~391,000 UGX/day (~USD 106/day)
Annualized	~USD 37,000/year
Recovered by measurement alone (added water fell 0.49pp)	~USD 11,000/year
Cost of a milk analyzer	~USD 500 (payback in days)

You do not need to test every delivery. Adulteration is concentrated: a small share of deliveries carries most of the water. Credible, selective testing at intake captures most of the leakage at trivial cost.

Simulation B. A quality premium that pays for itself.

Suppose you pay a component premium of about 100 UGX per liter per percentage point of butterfat. Moving average fat from 3.88 to 4.00 percent costs you roughly 12 UGX per liter in premium. Against that cost:

- Less water: 0.49 percentage points recovered at 1,200 UGX per liter is about 6 UGX per liter back immediately.
- More butterfat: a 0.12 percentage point gain is close to a 3 percent increase in recoverable fat, which raises butter, cheese, and powder yields on every liter processed.

On fat-driven products, the yield uplift typically covers the remainder of the premium, before you count the value of a higher, more consistent product grade and access to quality-sensitive export markets. The premium is, in round terms, self-financing.

Five recommendations

- 1. Pay for components, not for volume.** Replace flat per-liter pricing with payment that reflects measured butterfat and solids-not-fat (SNF). This is the single change that aligns what you pay with what you can actually process and sell.
- 2. Use one clear threshold, not a fine-grained gradient.** In our experiment, the continuous reward for marginal quality produced no detectable response. A single pass-or-fail bar did. Suppliers respond to a stake concentrated at one point, not to a smooth schedule they have to compute. Keep the rule simple: a premium above the bar, a discount or rejection below it.
- 3. Make the threshold dynamic and seasonal.** Raise the bar in the flush season, when milk is abundant and quality is cheap to supply, and relax it in the dry season, when milk is scarce. This approximates our finding that a standard only drives quality when it binds, and it protects your intake against the seasonal flood of watered milk during gluts.
- 4. Push the incentive to where quality is created, not just reshuffled.** A premium paid only to your immediate supplier can be satisfied by sorting: the trader hands you better milk by taking the best farmers and pushing the rest to another buyer. You get quality, but the chain's stock of good milk has not grown, and every processor ends up bidding for the same scarce supply. To expand quality rather than just capture it, the reading and the reward have to reach the farmers who actually set milk composition through feeding, hygiene, and handling. A quality-and-payment receipt sent to the producer (for example by short message service, SMS) also makes your grading credible and builds a loyal supply base that responds, which reduces disputes and side-selling.
- 5. Buy the cheapest quality win first: honesty.** Curbing adulteration requires no farmer investment and no behavior change on the production side, only credible testing at your intake gate. It is the fastest return in the chain. Sequence your investment accordingly: analyzers and transparent grading first, production-side premiums second.

Bonus recommendation. Lock in loyal quality supply with forward commitments. Offer guaranteed premiums or seasonal contracts to consistent high-quality suppliers. Relational contracts reduce the incentive to side-sell watered milk to whichever buyer is not testing today, and they stabilize both your volume and your grade.

Bottom line

Paying for quality is not a concession to farmers. It is how you stop subsidizing water, lift your processing yields, and secure a supply base that can reach premium and export markets. The technology is cheap, the quality already exists in the chain, and the first returns arrive within weeks. The only thing missing is a buyer willing to pay for what can be measured. Be that buyer first.